

Seasoned Data Scientist and Machine Learning Engineer with 8+ years of end-to-end AI product delivery. Proven ability to translate challenges into scalable, production-grade solutions that integrate with medical systems, cloud platforms, and embedded applications. 15+ publications ranging from rare disease prediction, bias mitigation, to deep learning on EKG.

Experience

- **Senior Machine Learning Engineer** October 2025 - Present
Visual Thought Labs San Francisco, CA
 - Developed ambient drug OCR in VA operating rooms, deployed on Nvidia Jetson at 77% lower hardware cost and 73% lower latency; built the SQL pipeline populating it, and HL7 system issuing results.
- **Senior Data Scientist** January 2022 - October 2025
UCSF San Francisco, CA
 - Built "Drifter," a data-drift monitoring tool used by 80 researchers across six hospitals; surfaced feature, label, and prediction drift that triggered timely ETL updates.
 - Directed a nation-wide matched-cohort study on traumatic-brain-injury outcomes, quantifying a 67% higher risk of malignant brain tumors (HR = 1.67) and publishing findings in JAMA Network Open.
- **Machine Learning Engineer** October 2021 - January 2022
Syntegra San Francisco, CA
 - Created a GenAI synthetic patient data generator that transformed structured EHR records into natural-language sentences, trained BERT models, and regenerated structured data, enabling safe data sharing for rare-case research.
- **Lead Data Scientist** December 2020 - October 2021
LifeBell AI Atlanta, GA (remote)
 - Deployed early sepsis detection system using XGBoost, with clinical explainer at Naval Medical Center San Diego.
 - Designed a Dynamic-Time-Warped Gaussian Mixture Model for phenotype discovery, surfacing sepsis-risk subgroups several hours before conventional detection.
- **Data Scientist** August 2018 - December 2020
UCSF San Francisco, CA
 - Developed deep-learning EKG classifier for pulmonary hypertension, achieving an AUC of 0.89 and early-diagnosis capability up to two years before conventional testing.
 - Led feasibility analyses that helped secure \$5M+ in research partnerships (J&J, Roche, Genentech).
- **Embedded Systems Engineer** May 2014 - August 2018
NASA Education & Public Outreach Rohnert Park, CA
 - Conceived/built/launched EdgeCube 10cm satellite, overseeing sensor integration, data logging, and radio telemetry.
 - Developed a STEM curriculum ("Learning By Making") that raised high-school science scores by 7.8% versus controls.

Education

- **MS in Computational and Mathematical Engineering** 2018
Stanford University Stanford, CA
- **BS Physics and Astronomy, and BA Mathematics** 2014
Sonoma State University Rohnert Park, CA
Graduated with Distinction in Physics, and Magna Cum Laude
- **AS in Mathematics** 2011
Santa Rosa Junior College Santa Rosa, CA

Publications

- T. D. Brender, N. Stumpf, K. Vossler, S. Kim, **H. Mills**, A. Lee, H. Y. Yang, T. Heintz, E. Espejo, J. Boscardin, et al. “How ICU clinicians document ‘Futility’: A 10-year analysis of critical care notes using natural language processing notes”. In: *Journal of Critical Care* 94 (2026), p. 155526.
- C. W. Cu, T. Heintz, N. Dundas, H. Farhan, **H. Mills**, J. W. Lee, C. Williams, A. W. Wallace, and J. Cobert. “Time and Motion Analysis of Controlled Substance Disposals: A Study of Workflows at a Single Center using Automated Dispensing Cabinets”. In: *Anesthesiology Open* 1.1 (2026), e0006.
- N. E. Dundas, T. Law, T. Brender, **H. Mills**, E. Espejo, T. A Heintz, A. W. Wallace, and J. Cobert. “All That Shines Is Not Gold: Maintaining Scientific Rigor When Evaluating, Interpreting, and Reviewing Studies Using Large Language Models”. In: *Anesthesiology* 144.2 (2026), pp. 272–288.
- E. Farrand, A. Chung, J. Joshua, H. Dong, **H. Mills**, A. Lee, M. Jeong, L. Radhakrishnan, O. Gologorskaya, and A. Butte. “Development and validation of a generalisable machine learning algorithm for identifying interstitial lung disease cohorts: a retrospective cohort study”. In: *Eclinicalmedicine* 93 (2026).
- C. Halabi, **H. Mills**, A. M. DiGiorgio, G. Schenk, S. Israni, C. Peek-Asa, and G. T. Manley. “Civilian Traumatic Brain Injury Is Associated with Longitudinally Increased Risk of PTSD, Suicidality, and Other Mental Health Diagnoses”. In: *Journal of Neurotrauma* (2026), p. 08977151251414145.
- C. W. Cu, N. E. Dundas, T. Heintz, Z. A. Sheikh, B. Alonso-Bermudez, J. Walker, A. Wooten, A. Badathala, A. Chapman, O. Ehie, K. Raghunathan, **H. Mills**, et al. “Validity of two subjective skin tone scales and its implications on healthcare model fairness”. In: *NPJ Digital Medicine* 8.1 (2025), p. 595.
- S. Marini, A. R. Alwakeal, **H. Mills**, J. D. Bernstock, A. Mashlah, M. T. Hassan, J. Gerstl, F. Radmanesh, G. Schenk, S. Israni, et al. “Traumatic brain injury and risk of malignant brain tumors in civilian populations”. In: *JAMA Network Open* 8.8 (2025), e2528850.
- G. Marquez-Grap, A. Leung, F. Orcales, A. Kranyak, C. Johnson, P. Smith, K. Haran, **H. Mills**, G. Schenk, A. Kahlon, et al. “Monitoring atopic dermatitis using mobile-app based photography and surveys”. In: *Journal of Dermatological treatment* 36.1 (2025), p. 2555983.
- A. Mashlah, S. Marini, **H. Mills**, T. Yahya, F. Radmanesh, J. Chalif, B. E. Zusman, J. D. Bernstock, M. H. Al Mansi, R. Grashow, et al. “Traumatic Spinal Cord Injury and Subsequent Risk of Developing Chronic Cardiovascular, Neurologic, Psychiatric, and Endocrine Disorders”. In: *JAMA Network Open* 8.11 (2025), e2541157.
- S. K. McGowan, M.-J. Corrales-Martinez, T. Brender, A. K. Smith, S. Kim, K. L. Harrison, **H. Mills**, A. Lee, D. Bamman, and J. Cobert. “Unclear Trajectory and Uncertain Benefit: Creating a Lexicon for Clinical Uncertainty in Patients with Critical or Advanced Illness Using a Delphi Consensus Process”. In: *Medical Decision Making* 45.1 (2025), pp. 34–44.
- S. Miramontes, U. Khan, S. Zimmerman, E. L. Ferguson, **H. Mills**, B. Oskotsky, E. Phelps, T. T. Oskotsky, A. J. Capra, M. Glymour, et al. “Diagnostic Reversion in Dementia Care: A Real-World Analysis of Mild Cognitive Impairment Diagnoses Following Dementia in a Large Electronic Medical Record System”. In: *medRxiv* (2025).
- H. Y. Yang, K. Raghunathan, E. Widera, S. Z. Pantilat, T. Brender, T. A. Heintz, E. Espejo, J. Boscardin, **H. Mills**, A. Lee, et al. “Lexical associations can characterize clinical documentation trends related to palliative care and metastatic cancer”. In: *Scientific Reports* 15.1 (2025), p. 17245.
- J. Cobert, E. Espejo, J. Boscardin, **H. Mills**, D. Ashana, K. Raghunathan, T. A. Heintz, A. C. Chapman, A. K. Smith, and S. Lee. “Variation in Mentions of Race and Ethnicity in Notes in Intensive Care Units Across a Health Care System”. In: *American Journal of Critical Care* 33.6 (2024), pp. 462–466.
- J. Cobert, **H. Mills**, A. Lee, O. Gologorskaya, E. Espejo, S. Y. Jeon, W. J. Boscardin, T. A. Heintz, C. J. Kennedy, D. C. Ashana, et al. “Measuring implicit bias in ICU notes using word-embedding neural network models”. In: *Chest* 165.6 (2024), pp. 1481–1490.
- C. Halabi, S. Izzy, A. DiGiorgio, **H. Mills**, F. Radmanesh, J. Yue, et al. *Traumatic Brain Injury and Risk of Incident Comorbidities. JAMA Netw Open* 7: e2450499–e2450499. 2024.
- S. Kim, **H. Mills**, T. Brender, S. McGowan, E. Widera, A. C. Chapman, K. L. Harrison, S. Lee, A. K. Smith, D. Bamman, et al. ““My Mom Is a Fighter”: A Qualitative Analysis of the Use of Combat Metaphors in ICU Clinician Notes”. In: *Chest* 166.5 (2024), pp. 1162–1172.

- A. Tang, K. Rankin, G. Ceron, S. Miramontes, **H. Mills**, J. Roger, B. Zeng, C. Nelson, K. Soman, S. Woldemariam, et al. *Leveraging electronic health records and knowledge networks for Alzheimer's disease prediction and sex-specific biological insights*. *Nat Aging* 4, 379–395. 2024.
- M. A. Aras, S. Abreau, **H. Mills**, L. Radhakrishnan, L. Klein, N. Mantri, B. Rubin, J. Barrios, C. Chehoud, E. Kogan, et al. “Electrocardiogram detection of pulmonary hypertension using deep learning”. In: *Journal of cardiac failure* 29.7 (2023), pp. 1017–1028.
- E. Farrand, O. Gologorskaya, **H. Mills**, L. Radhakrishnan, H. R. Collard, and A. J. Butte. “Machine-learning algorithm to improve cohort identification in interstitial lung disease”. In: *American Journal of Respiratory and Critical Care Medicine* 207.10 (2023), pp. 1398–1401.
- M. Mahendra, Y. Luo, **H. Mills**, G. Schenk, A. J. Butte, and R. A. Dudley. “Impact of different approaches to preparing notes for analysis with natural language processing on the performance of prediction models in intensive care”. In: *Critical care explorations* 3.6 (2021), e0450.

Awards

■ Hero Award (Tubbs Fire) Stanford University	2017
■ Departmental Fellowship Stanford University	2015
■ Provost Undergraduate Research Award Sonoma State University	2014
■ Future Faces of Physics Society of Physics Students	2014
■ Marsh W. White Award Society of Physics Students	2014
■ Sigma Pi Sigma Undergraduate Research Award Society of Physics Students	2014
■ Newkirk Student Assistantship Sonoma State University	2012
■ Doyle Scholarship Santa Rosa Junior College	2007

Certifications

■ ColumbiaX	
□ CSMM101x: Artificial Intelligence (2017)	□ CSMM102x: Machine Learning (2017)

Skills

■ Languages	Python, SQL, Julia, C, C++, JavaScript, TypeScript
■ ML Frameworks	PyTorch, TensorFlow, Scikit-Learn, XGBoost, CVXPY, RAPIDS, JAX, TensorRT, Ollama, HuggingFace, StatsModels, lifelines
■ Infrastructure & Compute	Spark, Spark-SQL, Hadoop, MPI, OpenMP, CUDA, AWS, GCP, Docker

Volunteering/Open Science/Open Source

■ Open Source	
□ pymargins (Author)	□ NDI (Contributer)
□ sklearn-onnx (Contributer)	□ pillow-jpegxl-plugin (Contributer)
■ Data Scientist Eclipse MegaMovie Project	2023 - 2025
■ Firefighter Sebastopol Fire Department	January 2017 - June 2019